

MODEL COMPARISON REPORT

House Price Prediction Using Machine Learning

Dataset: Ames Housing Dataset

Project Type: Regression Analysis

Evaluation Metrics: MAE, RMSE, R^2 Score

1. Introduction

1.1 Overview

After completing data preprocessing and exploratory data analysis, several machine learning regression algorithms were trained and evaluated to predict house prices. The objective of this stage was to compare the performance of different models and identify the most suitable algorithm for accurate house price prediction.

The comparison was performed using standard regression evaluation metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

2. Models Used

The following regression algorithms were implemented and evaluated:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor
- LightGBM Regressor

- CatBoost Regressor

Each model was trained using the same training dataset to ensure a fair comparison.

3. Model Evaluation Metrics

The following evaluation metrics were used:

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted house prices.

Lower MAE indicates better performance.

Root Mean Squared Error (RMSE)

RMSE measures the square root of the average squared prediction error.

Lower RMSE indicates more accurate predictions.

R² Score

The R² Score measures how well the model explains the variance in the target variable.

Higher R² Score indicates better performance.

4. Model Performance Comparison

Before optimization:

Model	MAE	RMSE	R2 Score
CatBoost	0.084051	0.130006	0.909429
Gradient Boosting	0.087975	0.132423	0.906030
LightGBM	0.092246	0.140955	0.893531
Random Forest	0.097504	0.147698	0.883101
XGBoost	0.108077	0.157842	0.866492

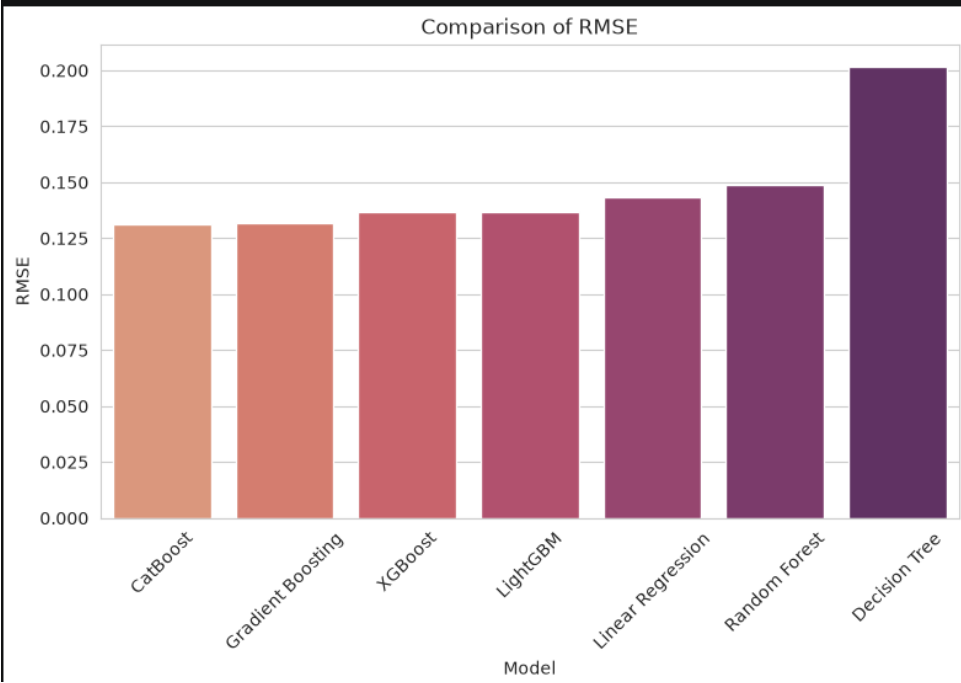
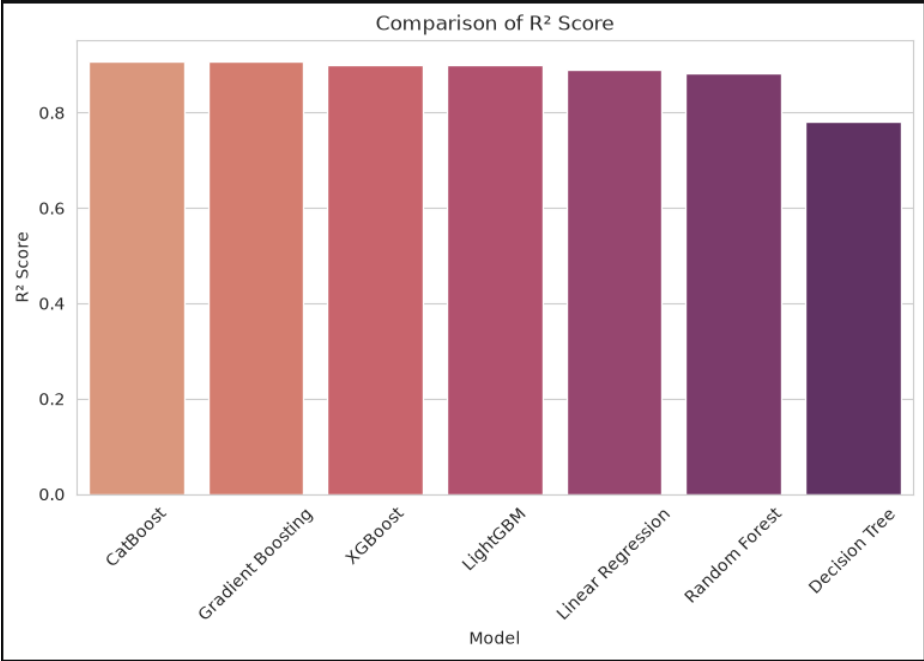
Linear Regression	0.093513	0.185011	0.816575
Decision Tree	0.141863	0.209322	0.765204

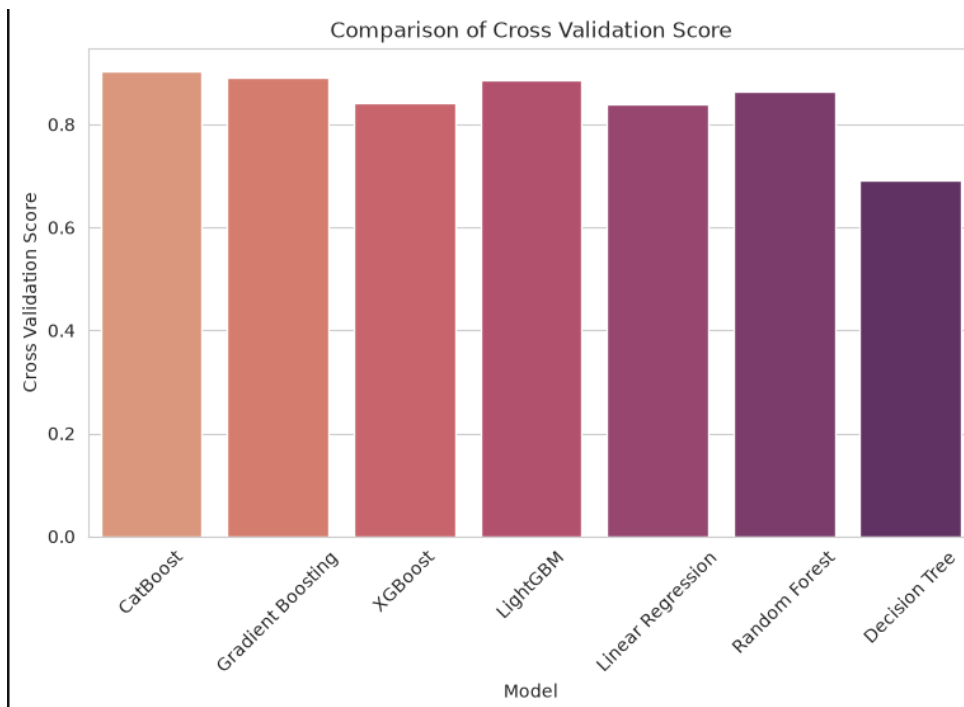
After Optimization:

Model	MAE	MSE	RMSE	R2 Score	Adjusted R2	CV Score
CatBoost	0.086152	0.017281	0.131456	0.907398	0.037596	0.903125
Gradient Boosting	0.086001	0.017344	0.131697	0.907057	0.034060	0.889995
XGBoost	0.091521	0.018659	0.136600	0.900009	-0.039196	0.841779
LightGBM	0.088333	0.018668	0.136631	0.899963	-0.039668	0.886280
Linear Regression	0.088336	0.020583	0.143468	0.889701	-0.146324	0.840067
Random Forest	0.097788	0.022090	0.148628	0.881624	-0.230266	0.863269
Decision Tree	0.137658	0.040653	0.201625	0.782153	-1.264056	0.692387

5. Performance Visualization

The comparison charts provide a clear visual representation of model performance





6. Model Analysis

The comparison results indicate noticeable differences in the prediction performance of the regression models.

Linear Regression

- Simple baseline model.
- Performs well on linear relationships.
- Unable to capture complex patterns in housing data.

Decision Tree

- Easy to interpret.
- High variance.
- More prone to overfitting.

Random Forest

- Better generalization than Decision Tree.

- Reduced overfitting.
- Improved prediction accuracy.

Gradient Boosting

- Strong predictive performance.
- Sequential learning improves accuracy.
- Handles complex relationships effectively.

XGBoost

- High computational efficiency.
- Excellent predictive capability.
- Performs well on structured datasets.

LightGBM

- Faster training.
- Efficient memory usage.
- Suitable for large datasets.

CatBoost

- Best overall performance.
- Handles categorical features effectively.
- Produced the highest prediction accuracy among all evaluated models.

7. Best Model Selection

The comparison results show that **CatBoost Regressor** achieved the best overall performance based on the evaluation metrics.

Reasons for Selecting CatBoost

- Highest R^2 Score.
- Lowest prediction error.

- Stable performance.
- Better generalization compared to other models.
- Suitable for structured tabular data.

8. Conclusion

The comparative analysis demonstrated that ensemble learning algorithms significantly outperformed traditional regression models for the house price prediction task.

Among all evaluated algorithms, **CatBoost Regressor** achieved the best balance between prediction accuracy and error minimization. Based on these results, CatBoost was selected for further optimization through cross-validation and hyperparameter tuning.